

Notice of Intent

The National Institute of Standards and Technology (NIST) Acquisition Management Division intends to negotiate with EXFO America Inc., located in Richardson, TX, on a sole source basis under the authority of FAR Subpart 13.106-1(b), soliciting from a single source, for Optical Fiber Automation Equipment.

Specific Requirements are as follows:

NIST will be purchasing a system of switches, shutters and attenuators from which to build an automated version for the optical power meter calibration setup. This automation equipment will be delivered to the Army which will replace the over 20-year-old equipment in current use as part of the Optical Fiber Modernization effort.

The following technical specifications are required for Optical Fiber Automation Equipment.

The Contractor shall deliver:

The Contractor shall provide a system that meets all technical specifications identified below. All items must be new. Used or remanufactured equipment will not be considered for award. Experimental, prototype, or custom items will not be considered. The use of “gray market” components are not authorized for sale in the U.S. by the Contractor is not acceptable. All line items shall be shipped in the original manufacturer’s packaging and include all original documentation and software, when applicable.

Line Item 0001:

Description: Optical Fiber Automation Equipment

Quantity: 1

A. Technical Specifications for entire system

1. A compact system such as a mainframe with the following subcomponents
 - a. 3 x 1 fiber switch
 - b. 2 x 1 fiber switch
 - c. 2 fiber optical shutters
 - d. 2 optical attenuators.

The technical specifications for each of the subcomponents will be described separately.

2. The system must be optical fiber based, with SMF-28 fiber and FC/APC connections.
3. The system should be addressed by a single computer connection via GPIB
4. It must be possible to control the instrument via custom LabVIEW or Python software.
5. The sum of insertion losses of all subcomponents must be less than 8 dB. This can be achieved with for example < 1.4 dB insertion loss for every subcomponent, or by having some subcomponents with higher and some with lower insertion losses, summing to less than 8 dB.
6. All subcomponents must tolerate 50 mW input power.

B. Technical specifications for subcomponent A, 3x1 fiber switch

1. FC/APC input and output for SMF-28 fiber

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2. The switch must connect any of at least three inputs to a single output. Larger numbers of inputs or outputs are acceptable, however.
3. The switch must operate over a range from 1260-1650 nm.
4. Must be computer controllable via GPIB.

C. Technical specifications for subcomponent B, 2x1 fiber switch

1. FC/APC input and output for SMF-28 fiber
2. The switch must connect any of at least two inputs to a single output. Larger numbers of inputs or outputs are acceptable, however.
3. The switch must operate over a range from 1260-1650 nm.
4. The switch must be repeatable to better than 0.1% over a period of an hour.
5. Must be computer controllable via GPIB.

D. Technical specifications for subcomponent C, optical shutter

1. FC/APC input and output for SMF-28 fiber
2. The shutter must block the light with better than 80 dB rejection in the wavelength range from 1260 nm – 1650 nm.
3. Tolerate 10 mW input power.
4. Must be computer controllable via GPIB.

E. Technical specifications for subcomponent D, optical attenuator

1. FC/APC input and output for SMF-28 fiber
2. Operate at 1310 and 1550 nm with attenuations from 0 to 75 dB possible.
3. Must be computer controllable via GPIB.
4. The total attenuation could be achieved (for example) with two attenuator modules in series if necessary.

NAICS Code is 334516 – Analytical Laboratory Instrument Manufacturing, with a Size Standard of 1,000 employees. NIST anticipates negotiating and awarding a firm-fixed-price purchase order to EXFO America Inc. for this requirement.

Interested parties that can demonstrate they could satisfy the requirement listed above for NIST must clearly and unambiguously identify their capability to do so in writing on or before the response date for this notice. This notice of intent is not a solicitation. Information submitted in response to this notice will be used solely to determine whether competitive procedures could be used for this acquisition. If competitive procedures are not used it is estimated that an award will be issued by 4/28/2025. Any questions regarding this notice must be submitted in writing via email to matthew.delgado@nist.gov. All responses to this notice of intent must be submitted to matthew.delgado@nist.gov no later than the date and time provided herein this posting.

JUSTIFICATION FOR OTHER THAN FULL AND OPEN COMPETITION:

This purchase must be sole-source to allow for simple integration of the Automation Equipment into the Army Modernization system. NIST has already developed and tested a modernization system (using NIST-provided modernization funds) that operates at a low-uncertainty and with

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high repeatability. The NIST system utilizes a specific brand of switches/attenuators/shutters (Automation Equipment) that are very low-loss, highly repeatable, and easily mountable to a rack mount for integration with other parts of the system. Additionally, NIST has developed software that communicates with the automation equipment via GPIB. Many other vendors of Automation Equipment do not include GPIB communication in their products. If another vendor is purchased, there will be significant software development needed and possible reduced functionality of the system. Also, the system may not be as repeatable or low-loss. Further, procuring a different system would cost the government significantly in terms of additional staff time and materials.